Projective Temporal Ordering and Cumulative Projected Capacity: Numerical Evidence from the Spectral Admissibility Cascade

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Abstract

We identify a monotone proxy for temporal ordering within the Cosmochrony spectral admissibility programme. The natural observable along the BFS cascade is not the residual capacity $\sigma_{\text{pair}}(n)$, which decays by construction, but the cumulative projected capacity $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$, which measures the spectral structure already projected at depth n. For $q \in \{29, 61, 101, 151, 211\}$, $I(n)$ is found to be strictly non-decreasing at every BFS step, for every conjugate pair $(c, q-c)$, with zero physically meaningful regressions observed across 274 pairs and 360 steps after burn-in. The temporal ordering is not a dynamics on the substrate χ, but an induced ordering relation on admissible projections, compatible with the observable rank stability established in O24 [6]. The saturation plateau of $I(n)$ is interpreted as Born–Infeld saturation at the pair level.

Keywords. non-injective projection; projective temporal ordering; spectral admissibility; cumulative projected capacity; one-way activation; Weil representation; Heisenberg group; Gram–Schmidt dynamics; emergent time

1 Introduction

The status of time in fundamental physics remains unsettled. In classical mechanics, time appears as an external parameter; in general relativity, it is incorporated into spacetime geometry; in quantum theory, it remains an external evolution parameter with no operator counterpart. Despite their empirical success, these frameworks do not provide a unified account of the origin of temporal ordering.

In thermodynamic approaches, the arrow of time is associated with entropy increase. However, such accounts presuppose probabilistic structure and coarse-graining, and do not explain why a preferred ordering exists at the level of fundamental description. More recent approaches attempt to derive time from correlations, quantum entanglement, or relational observables, but typically retain an underlying notion of evolution or state update.

The Cosmochrony framework adopts a different starting point [3]. The relational substrate χ is static, and no fundamental time parameter is assumed. Observable dynamics arises instead from an iterated reprojection sequence

$$U_{n+1} = \Pi \circ \sigma(U_n), \tag{1}$$

where Π is a fixed non-injective projection and $\sigma(U_n) \in \Pi^{-1}(U_n)$ selects an admissible configuration within the fibre compatible with the current effective state. In this setting, the notion of time cannot be identified with an evolution of χ, but must emerge from the structure of admissible projections itself.

This raises a precise question: does there exist a scalar observable, defined intrinsically along the admissibility cascade, that induces a consistent temporal ordering without introducing additional structure?

The spectral admissibility programme provides a natural testing ground for this question. For each prime q and each conjugate pair $(c, q-c)$, the pipeline produces a sequence of observables $\sigma_{\text{pair}}(n)$ indexed by BFS depth n, interpreted as residual projective capacity [4]. While $\sigma_{\text{pair}}(n)$ is not suitable as a temporal parameter due to its decreasing behaviour, its complement

$$I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n) \tag{2}$$

measures the cumulative projected capacity, i.e. the amount of spectral structure stably realised up to depth n.

The central result of this work is that $I(n)$ is strictly non-decreasing along all tested admissible trajectories, across all conjugate pairs and all primes $q \in \{29, 61, 101, 151, 211\}$. This monotonicity is not imposed but observed numerically at each step of the cascade, with no physically meaningful regression.

We interpret this property as defining a *projective temporal ordering*: time is not a parameter of the substrate, but a monotone ordering induced

by the accumulation of admissible spectral structure under a non-injective projection. This ordering is compatible with the stability of observable rank established in O24 [6] and introduces no additional dynamical degrees of freedom.

This perspective suggests that temporal ordering may arise as a structural consequence of non-injectivity, rather than as a fundamental ingredient of the theory. The cumulative observable $I(n)$ provides a concrete and computable realisation of this idea within the spectral admissibility cascade.

Relation to measurement-induced arrows of time. Recent work shows that the arrow of time in monitored quantum systems depends explicitly on the measurement protocol and can be statistically reversed by engineered Hamiltonians combined with feedback [10]. Within the present framework, this behaviour admits a direct structural interpretation.

The monotonicity of the cumulative projected capacity $I(n)$ reflects the information loss induced by the non-injective projection Π. A feedback-controlled reversal does not contradict this structure: it compensates for this loss by injecting additional external information, effectively reconstructing part of the preimage of Π. This supports the interpretation that temporal asymmetry is not a primitive feature of the substrate, but an emergent consequence of projective selection under admissibility constraints.

1.1 Conditional Quantum Models and the Limits of Reversible Emergence

Several approaches to the emergence of time in quantum theory rely on conditional structures rather than fundamental evolution. A notable example is the Page–Wootters construction, in which a global stationary state $|\Psi\rangle$ on a composite Hilbert space $\mathcal{H}_C \otimes \mathcal{H}_S$ induces an apparent evolution via conditional states

$$|\psi(t)\rangle := \langle t|\Psi\rangle.$$

This construction can be interpreted as a family of conditional projections

$$\Pi_t(\Psi) := \langle t|\Psi\rangle,$$

which generate an ordering parameter t without introducing an external time.

From the present perspective, this mechanism shares a structural similarity with the projective cascade: both frameworks extract an effective sequence of states from a higher-level configuration via projection.

However, a crucial difference arises. In conditional quantum models, the projection is not structurally non-injective in the physical sense: the global state $|\Psi\rangle$ is assumed to be fully recoverable, and the apparent evolution

remains reversible. The ordering parameter t labels different conditionings but does not define an intrinsically oriented sequence.

By contrast, in the projective admissibility framework, the projection Π is structurally non-injective and induces irreversible information loss at each step. The cumulative projected capacity

$$I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$$

quantifies this loss as a monotone accumulation of realised structure.

The observed strict monotonicity of $I(n)$ therefore provides a structural criterion distinguishing genuine temporal ordering from reversible conditional parametrisations: only when projection entails irreversible compression does an intrinsic arrow of time emerge.

In this sense, conditional constructions such as Page–Wootters can be understood as reversible limits of the more general non-injective projection framework, in which ordering exists but lacks orientation.

Relation to conditional quantum models. Relational approaches to the emergence of time, such as the Page–Wootters construction [12], recover an effective temporal ordering by conditioning on a clock subsystem within a globally stationary state. From the present perspective, such constructions involve a projection structure whose non-injective character is not exploited: the effective evolution is reconstructed without an associated cumulative loss of distinguishable information. The projective temporal ordering identified here differs in this respect, being tied to the monotone accumulation of projected capacity along the admissible cascade rather than to the choice of a conditioning sector.

2 The Two Variables and Why σ_{pair} Is Not the Right One

The observable $\sigma_{\text{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$ measures the residual projective capacity of the conjugate pair at depth n: it counts, normalised by q, the fraction of Weil fingerprint vectors that remain linearly independent of the Gram–Schmidt span built over the preceding shells. By construction, $\sigma_{\text{pair}}(n)$ decays as n grows and cannot serve as a temporal proxy, since a decreasing quantity does not define an ordering compatible with the accumulation of structure. The correct variable is its complement (2), which we now define formally.

Definition 1 (Cumulative projected capacity)**.** *Let $\sigma_{\text{pair}}(n)$ denote the canonical pair observable [4] at BFS depth n. The* cumulative projected capacity *at depth n is*

$$I(n) \;:=\; \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n). \tag{3}$$

$I(n)$ measures the spectral structure stably projected up to depth n: it starts at zero and accumulates as the cascade progresses, with $I(0) = 0$ and $I(n) \to \sigma_{\text{pair}}(0)$ as $n \to n_{\text{sat}}$, where n_{sat} denotes the saturation depth at which $\sigma_{\text{pair}}(n) \approx 0$.

Remark 1. *$I(n)$ is non-decreasing if and only if $\sigma_{\text{pair}}(n)$ is non-increasing. While this is expected under a dissipative interpretation of the BFS cascade, the present work establishes it empirically without relying on a formal dissipation theorem: the monotonicity of $I(n)$ is observed directly, step by step and pair by pair, across all tested primes.*

Independence from the BFS parametrisation. Although $I(n)$ is indexed by BFS depth, its definition does not depend on the combinatorial parametrisation of the cascade but on the cumulative spectral projection. The monotonicity observed here is therefore not an artefact of BFS ordering, but reflects an intrinsic property of admissible projection: the Gram–Schmidt span can only grow within the canonical Gram–Schmidt construction used in the admissibility pipeline.

3 Temporal Ordering as a Projective Relation

Order versus dynamics. The temporal structure proposed here is not a dynamics on the substrate Ω and not an evolution of χ. It is an induced ordering relation on the set of admissible projectable configurations. The effective time parameter arises as a monotone ordering on admissible projections, defined by the progressive saturation of the Born–Infeld constraint under Π. Time is therefore neither a parameter of χ nor an independent structure on the observable space $\mathcal{O}$, but a relational ordering induced by the projective admissibility structure.

Temporal ordering and non-injectivity. Because Π is non-injective [2], a given observable state $o \in \mathcal{O}$ corresponds to a fibre $\Pi^{-1}(o)$ containing multiple underlying configurations. Temporal ordering does not correspond to motion within Ω, but to a change in the admissible selection $\sigma(U_n) \in \Pi^{-1}(U_n)$ that defines the next effective state $U_{n+1} = \Pi \circ \sigma(U_n)$. Successive steps correspond to distinct admissible realisations drawn from different regions of the fibres. Time is therefore an ordering of admissible selections within a non-injective structure, rather than an evolution of the underlying configuration χ.

Conversely, if Π were injective, each observable would correspond to a unique underlying configuration and no ordering ambiguity would arise. The existence of temporal ordering is thus a direct consequence of non-injectivity: it reflects the structural impossibility of identifying a unique underlying configuration for each observable state.

Proposition 1 (Projective temporal ordering — numerical form)**.** *Define an ordering on effective states along the BFS cascade by*

$$U_n \prec U_{n+1} \quad \Longleftrightarrow \quad I(U_n) \leq I(U_{n+1}), \tag{4}$$

where n is the BFS shell depth, an ordering parameter on projected configurations that does not represent a physical time variable at the substrate level. For $q \in \{29, 61, 101, 151, 211\}$, this ordering is strictly observed numerically along the tested BFS trajectories: $I(n)$ is strictly non-decreasing at every step, for every conjugate pair $(c, q-c)$, with no physically meaningful exception.

Remark 2 (Status)**.** *This is numerical evidence, not a proof. The result establishes the ordering empirically along the tested trajectories and does not exclude the possibility of exceptions at larger q or along non-BFS paths. The one-way activation property—that modes, once stably projected, do not deactivate—remains to be established analytically.*

Remark 3 (State machine structure of the admissible cascade)**.** *The admissible neighbourhood $\mathcal{A}(U_n)$ gives the projective cascade the structure of an oriented state machine: nodes are admissible projections, edges are admissible transitions, and a physical history is a directed path. When $|\mathcal{A}(U_n)| = 1$, the successor is uniquely determined by the admissibility structure — no external dynamical law is required. When branching occurs ($|\mathcal{A}(U_n)| > 1$), the relative admissibility weights of the branches play the role of quantum probabilities. In both cases, the index n acquires its temporal meaning entirely from the orientation of the admissibility graph; $I(n)$ measures cumulative path length in the spectral sense, not elapsed time with respect to any pre-existing background. The dynamics of the framework is the oriented topology of this graph, not a law superimposed upon it.*

Moreover, the first transition $U_0 \to U_1$ requires no external trigger. The projective incompleteness result of [3] establishes that any admissible initial projection Π_0 necessarily leaves a non-trivial residue, because a projection with trivial kernel would require the simultaneous resolution of non-commuting generators — which is structurally forbidden by the Heisenberg structure of the admissible fibre. The admissible neighbourhood $\mathcal{A}(U_0)$ is therefore guaranteed to be non-empty: the ordering starts not as a postulate but as a structural necessity.

Compatibility with observable rank stability. The temporal ordering introduced here does not enlarge the observable rank. It acts entirely at the level of admissible realisations within the fixed observable rank established in O23 [5] and stabilised in O24 [6]. In particular, temporal progression does not enlarge $\dim_{\mathbb{R}}(\mathrm{Im}\,\Pi \cap \mathrm{Ntrl})$, but only reorders admissible configurations within this fixed structure. The ordering is therefore compatible with the vertical non-injectivity theorem and introduces no hidden dynamics.

4 Protocol and Data

Computations use the `o25_paired_pipeline.py` pipeline with parameters `M=50` independent block samples per pair and `bfs_frac=0.99` (full graph exploration). Monotonicity of $I(n)$ is tested by the `o_stability_diagnostic.py` script, which verifies, for each prime and each pair, that no step $I(n{+}1) < I(n)$ occurs after the burn-in depth $n_{\text{burn}} = 2$.

Table 1: Monotonicity and stability of $I(n)$ per prime. "Steps" counts BFS depths after burn-in. "Mean mono" reports whether the pair-averaged $I(n)$ is strictly non-decreasing. "Regressions" counts pairs with at least one physically meaningful decrease in I (threshold: relative decrease $> 10^{-6}$ of the plateau value).

q	pairs	steps	mean mono	regressions	mono frac
29	14	25	YES	0	1.0000
61	30	55	YES	0	1.0000
101	50	92	YES	0	1.0000
151	75	138	YES	0*	1.0000*
211	105	50	YES	0	1.0000

*At $q = 151$, 70 pairs exhibit floating-point fluctuations of order 10^{-10}–10^{-13} on the saturation plateau, where $I/I_{\max} = 1.000000$. These are $\sim$6 orders of magnitude below the mean positive step size of 4×10^{-2} and are not counted as physical regressions.

5 Interpretation

One-way activation. The zero-regression result means the admissible BFS cascade behaves as a one-way activation process in spectral space. Once a mode contributes stably to the Gram–Schmidt span, it does not leave: the spectral structure accumulated at depth n is preserved at all subsequent depths. One-way activation is a property of *spectral modes*, not of the coherence of projected configurations: a coherent combination of modes — corresponding to an unstable particle — can lose its admissible coherence and reorganize into simpler configurations without any mode leaving V_n. This one-way activation mechanism provides a structural form of irreversibility that does not rely on probabilistic or thermodynamic arguments, but follows from the admissible accumulation of projected spectral structure. It is the condition of possibility for the identification of $I(n)$ as a temporal proxy: an ordering defined by I is well-posed only if I never reverses.

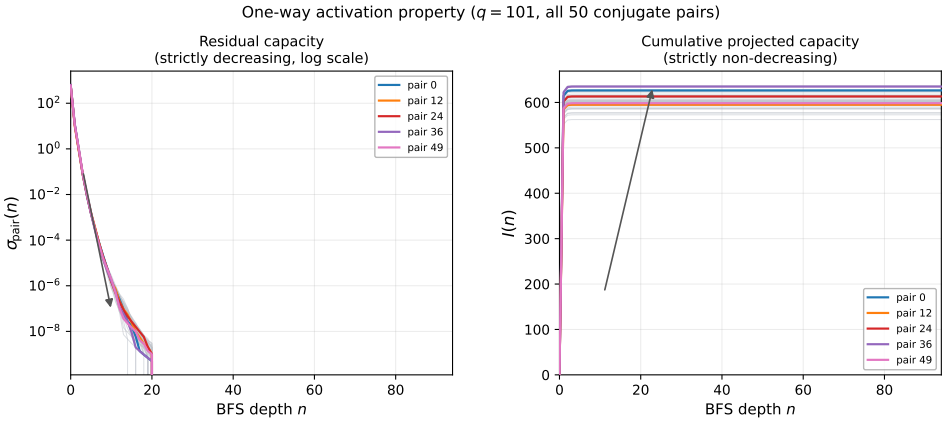


Figure 1: Left: residual capacity $\sigma_{\text{pair}}(n)$, strictly decreasing (log scale). Right: cumulative projected capacity $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$, strictly non-decreasing. Both panels show all 50 conjugate pairs at $q = 101$. The two observables are exact complements, illustrating the one-way activation property: spectral modes, once projected, are never lost from the Gram–Schmidt span (coherent configurations may nonetheless reorganize).

Non-arbitrariness of $I(n)$. The choice of $I(n)$ as a temporal proxy is not arbitrary among monotone functions: it is uniquely determined within the present framework as the cumulative quantity associated with admissible spectral projection, and therefore carries direct physical meaning. It is bounded above by $\sigma_{\text{pair}}(0)$ (the Born–Infeld capacity of the pair) and its increments are exactly the newly projected spectral structure at each BFS step. The cumulative observable $I(n)$ is moreover expected to reflect the scaling properties encoded in the capacity exponent δ_{pair} [4], since both quantify the rate at which admissible spectral structure is resolved along the cascade; in particular, a larger value of δ_{pair} is expected to correspond to a faster accumulation of $I(n)$. Establishing this relation quantitatively remains an open problem (Question 3 below).

Information-theoretic interpretation. The cumulative projected capacity $I(n)$ admits a natural interpretation in information-theoretic terms. Since Π is non-injective [2], distinct admissible configurations at depth $n-1$ may map to the same observable state at depth n, implying a reduction of distinguishable information at the observable level. The quantity $I(n)$ measures the spectral structure stably projected up to depth n and is therefore compatible with an interpretation as a cumulative proxy for the information loss induced by Π. No strict identification with a projection entropy is assumed here; the observed monotonicity of $I(n)$ nonetheless supports the view that the projective temporal ordering reflects the progressive reduction

of distinguishable information along the admissible cascade.

Saturation plateau and Born–Infeld bound. $I(n)$ reaches its maximum $\sigma_{\mathrm{pair}}(0)$ when the residual capacity is exhausted. This plateau is interpreted as Born–Infeld saturation at the pair level: once the projective flux budget of the conjugate pair is fully consumed, no additional spectral structure can be resolved within the effective description. Saturation is irreversible in all tested cases; this is consistent with the bounded-flux constraint of the Cosmochrony framework [3] but is not derived from it here.

Candidate arrow of time. The strict monotonicity of $I(n)$ provides a candidate ordering compatible with an arrow of time, without invoking thermodynamic entropy or a background temporal coordinate. The ordering is projective: it reflects the irreversible accumulation of stably activated spectral structure along the admissible cascade. This provides a candidate identification of the temporal direction with the direction of increasing information loss induced by the non-injective projection Π, as quantified by the monotone accumulation of $I(n)$. Whether this identification holds within the full framework [3] is left as an open question; the present result establishes only its monotone and stable character along the tested BFS trajectories for $q \in \{29, 61, 101, 151, 211\}$.

Relation to standard time-dependent evolution. The present construction does not assume a fundamental time parameter. Temporal ordering is instead defined by the monotone accumulation of projected spectral capacity along the admissible cascade. In this reading, time-dependent evolution equations such as the Schrödinger equation [14] are naturally understood as effective descriptions formulated within a pre-existing temporal ordering, rather than as fundamental dynamical laws from which time itself originates. This perspective connects with longstanding discussions of the problem of time in quantum gravity [11, 1], where the absence of an external time parameter in the Wheeler–DeWitt equation [9] motivates relational and emergent approaches to temporal ordering [12, 13, 8]. This perspective also suggests a parallel with the distinction between time-dependent and time-independent formulations in quantum theory: while the time-dependent Schrödinger equation describes evolution with respect to an external parameter, the present construction is closer in spirit to a spectral characterization, in which the relevant structure is encoded in the set of admissible modes rather than in a fundamental time evolution. Whether the projective ordering identified here is compatible with such effective descriptions in appropriate limits remains an open question.

Formal statement and what remains to be proved

The numerical results above support the following conjecture.

Conjecture 1 (One-way activation)**.** *For all admissible primes q, all conjugate pairs $(c, q-c)$, and all $n \geq 1$:*

$$\sigma_{\text{pair}}(n) \;\leq\; \sigma_{\text{pair}}(n-1). \tag{5}$$

Three structural facts are immediate from the Gram–Schmidt construction. Let $V_n = \text{span}\{f_\omega : \omega \in S_0 \cup \cdots \cup S_n\}$ denote the cumulative span of Weil fingerprint vectors up to shell n. Then:

- $V_n \subseteq V_{n+1}$ (the span is non-decreasing),
- $\dim V_n \leq q$ (the ambient space has finite dimension q),
- $\delta r_n = \dim V_n - \dim V_{n-1} \geq 0$ (the rank gain is non-negative).

These facts imply that the cumulative rank $\dim V_n$ is non-decreasing and that $\sigma_{\text{pair}}(n) \to 0$ asymptotically as $\dim V_n \to q$. They do not, however, imply the stepwise monotonicity of $\sigma_{\text{pair}}(n)$, which is the content of Conjecture 1. However, they do not imply Conjecture 1: cumulative rank monotonicity alone does not control the shellwise ratio

$$\sigma_c(n) \;=\; \frac{\delta r_n}{|S_n|}, \tag{6}$$

since both numerator and denominator may grow with n. The conjecture requires, in addition, control of the efficiency with which successive BFS shells generate new independent Weil fingerprints relative to shell size.

The numerical results of the present paper show that this efficiency decreases monotonically throughout the tested range. A proof would require quantitative control of shellwise spectral saturation in the Weil sector of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, which we leave as an open problem (Question 2 in Section 6).

6 Open Questions

1. Does strict monotonicity of $I(n)$ hold for all admissible primes q, or are there exceptions at large q? The extension to $q = 211$ (105 pairs, 50 steps) confirms strict monotonicity with zero regressions (Table 1). Extension to $q \in \{307, 401\}$ is pending.

2. Can the one-way activation property—$\sigma_{\text{pair}}(n)$ non-increasing along the full cascade—be proved analytically from the structure of the BFS expansion and the admissibility constraints of χ?

3. What is the precise relationship between $I(n)$ and the capacity exponent δ_{pair} [4]? Is $I(n)$ sensitive to the value of δ_{pair} or only to its sign?

4. Does the saturation depth (the n at which $I(n)$ plateaus) carry physical meaning independent of q?

5. What is the precise relationship between BFS shell depth and effective resolution-cell entry? Preliminary numerical evidence suggests that the characteristic shell n^* at which $\sigma_{\text{pair}}(n)$ first undergoes a qualitative knee increases with q over the tested cases $q \in \{29, 61, 101\}$, with approximate values $n^*(29) \approx 4$, $n^*(61) \approx 6$, $n^*(101) \approx 8$. This suggests that BFS shells do not define resolution cells, but may reveal the depth at which such cells become effective. The structural status, scaling law, and analytical derivation of this relation are left open.

6. Does the characteristic shell $n^*(q)$ of Question 5 coincide with, or bound, the *locking depth* $n_c(q)$ at which the BFS admissibility structure transitions from a locally dominated regime to a universally dominated one? The companion paper Q10 [7] establishes that the horizontal metric coefficient $A_H(q)$ satisfies $A_H(q) \to 2$ as $q \to \infty$, where $2 = C_{\text{su}(2)}|_{j=1}$ is the Casimir eigenvalue on $\text{Sym}^2(V_\rho)$. At $q = 211$, the anomalous value $A_H(211) = 2.022$ (against the monotone trend at $q \leq 151$) is attributed in Q10 to insufficient BFS coverage at bfs_frac $= 0.11$, consistent with $n_c(211)$ exceeding the tested window at that fraction. This suggests that $n_c(q)$ is precisely the depth below which finite-coverage artefacts prevent the su(2)-isotropic structure from being resolved — analogous to the knee $n^*(q)$ at which σ_{pair} changes regime. Whether $n^*(q) \sim n_c(q)$, and whether $n_c(q)$ can be derived analytically from the admissibility structure of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, are open questions whose resolution would connect the spectral ordering of the present paper to the effective geometry results of [7].

7 Conclusion

The observable $\sigma_{\text{pair}}(n)$ measures residual projective capacity and decays along the BFS cascade; its complement $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ measures cumulative projected capacity and grows. Across $q \in \{29, 61, 101, 151, 211\}$, covering 274 pairs and 360 post-burn-in steps, $I(n)$ is strictly non-decreasing at every BFS step and for every conjugate pair, with no physically meaningful regression. This establishes $I(n)$ as a robust monotone proxy for temporal ordering along the admissible spectral cascade. The ordering is not a dynamics on χ but a relational structure induced by the non-injective projection Π: it reorders admissible fibre selections within the fixed observable rank of O24 [6], without introducing additional observable degrees of freedom. The one-way activation property observed here is a numerical result; its analytical derivation from the admissibility constraints of Π and the scaling of $n_1(q)/q$ [4] remains the central open direction. The monotonic accumulation of $I(n)$ is moreover compatible with an interpretation as cumulative information loss under non-injective projection, identifying the one-way activation property as the spectral signature of structural irreversibility. These results support the view that temporal ordering is an emergent consequence of admissible projection structure, rather than a primitive ingredient of physical theory.

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